Computer Vision — Project 5 · A.A. 2025/2026

Parameter-Efficient Few-Step Distillation of Diffusion Models via LoRA

Lightweight Diffusion Models for Resource-Constrained Environments

Generated samples · CIFAR-10

Claudia Giuseppini 2069387 **Benedetta Conti** 2045621 **Fabio Falconi** 2048079

Professor: **Irene Amerini**

Dip. di Ingegneria Informatica, Automatica e Gestionale — Sapienza Università di Roma

Outline

Problem Statement

DDPMs generate images by removing noise step by step and they need up to **1000 sequential network evaluations** per image. This is slow and expensive on limited hardware.

DDIM speeds this up by taking fewer, larger jumps but each jump adds **discretization error** that accumulates, so quality collapses at very few steps.

Our goal

- Reduce sampling time while keeping generation quality
- Fix the error that hurts few-step DDIM, not the per-step prediction
- Keep the extra training cheap enough for limited GPU

~1000

sequential steps for one DDPM image

325

FID of plain DDIM at a single step

0.1%

of weights we will train to fix this

State of the Art

DDPM

Denoising Diffusion Probabilistic Models — our baseline generator.

Limitation: ~1000 sequential steps → very slow sampling.

DDIM

Deterministic sampler that skips levels, enabling fewer steps.

Limitation: quality collapses under strong step reduction.

Distillation

Trains a student to reproduce a teacher in fewer steps.

Limitation: re-trains the whole network at each stage; very costly.

LoRA

Low-rank adapters for cheap adaptation of a frozen model.

Enables distillation with trajectory matching at ~0.1% cost.

Distillation can be used in State-of-the-Art in some of the references. LoRA is the State-of-the-Art for distillation in LLMs

Proposed Method 1

Baseline Improvements

Our first implementation follows the basic DDPM setup. This had difficulties producing clear images with resources we had (Kaggle/Colab).

Therefore, we decided to add some optimization to improve the baseline efficiently.

EMA – Exponential Moving Average

- We keep a second averaged copy of the weights
- Because the optimizer makes the weights oscillate around a good configuration this average is more stable .
- This also produces better images at no extra cost.

Mixed Precision

- Some computations are done in reduced (16-bit) precision which makes training faster and lighter on memory.
- There is no visible loss of quality.

Data Augmentation

- We apply a random horizontal flip to the training images
- A mirrored dataset image is still a plausible image, so this enriches the data without distorting its distribution, unlike crops and rotation.

Proposed Method 2

Parameter-efficient few-step distillation via LoRA

- Freeze the trained DDPM (EMA weights) as the teacher, add small low-rank adapters into the attention layers as the student.
- The teacher refines a state with m small DDIM sub-steps, the student must reach the same point in a single step.
- This teaches the student to compensate the error of large jumps: the student achieves few steps without the DDIM drop.
- Only A and B (adapters) are trained (start B = 0 ⇒ student = teacher); at inference they merge back: W′ = W + BA, no overhead.

TEACHER · frozen DDPM (EMA)

x_t → m small DDIM sub-steps → x_t' (target)

careful multi-step path

must match

STUDENT · DDPM + LoRA adapter

x_t → ONE DDIM step → x_t' (student)

one large jump, error compensated

Distillation objective

$$L = \| x_{\text{student}} - x_{\text{teacher}} \|^2 + \lambda \| \varepsilon_\theta - \varepsilon_\varphi \|^2$$

trajectory matching + noise-consistency (stabiliser)

16,384 trainable params · **0.107%** of 15.3M

Adapters merge into the weights

Experimental Setup

Model

- Attention U-Net, base_channels = 64, time_dim = 256
- ≈ 15.3M parameters; $T = 1000$ diffusion steps
- Linear noise schedule; ε-prediction objective
- Self-attention + LoRA

Training

- AdamW, batch size 128, 80 epochs (baseline)
- EMA teacher, AMP mixed precision, grad clipping
- Distillation: one student per $K \in \{8,4,2,1\}$, 10 epochs
- Only LoRA adapters trained (16,384 params)

Dataset – CIFAR 10

- 60,000 colour images, 32×32, 3 channels (RGB), across 10 classes.
- Split : 50,000 train / 10,000 test
- Pre-processing: pixels rescaled, so the data is zero-centered to match the zero-centered Gaussian noise of the diffusion process.

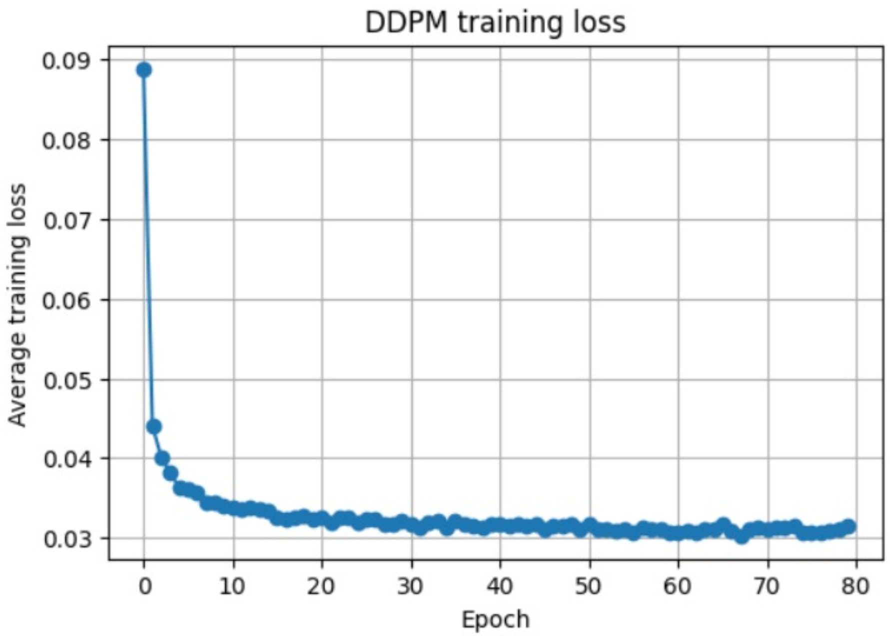
Hardware & reproducibility

- Single GPU — NVIDIA T4 (Kaggle)
- All metrics on 10,000 generated images
- Fixed seed = 42, deterministic algorithms
- Shared seeded noise bank → fair comparisons

Model Evaluation – Baseline

METRICS

- FID — image fidelity
- Inception Score — sharpness
- Precision & Recall — fidelity vs diversity



DDPM loss stabilizes after 40 epochs

Sampler	Steps	FID ↓	IS ↑	Precision	Recall	s/img	Speed-up†
DDPM (ref)	1000	67.3*	6.26	0.847	0.381	0.674	1×
DDIM	100	29.33	7.23	0.751	0.326	0.052	12.8×
DDIM	50	29.79	7.32	0.777	0.306	0.026	25.2×
DDIM	20	32.43	7.32	0.834	0.243	0.011	60.5×
DDIM	10	40.89	7.02	0.893	0.141	0.006	112.9×
DDIM	8	47.12	6.83	0.903	0.102	0.005	136.5×
DDIM	4	103.28	5.29	0.853	0.015	0.003	242.4×
DDIM	2	188.84	3.73	0.401	0.001	0.002	∼360×
DDIM	1	325.11	1.56	0.005	0.000	0.001	∼510×

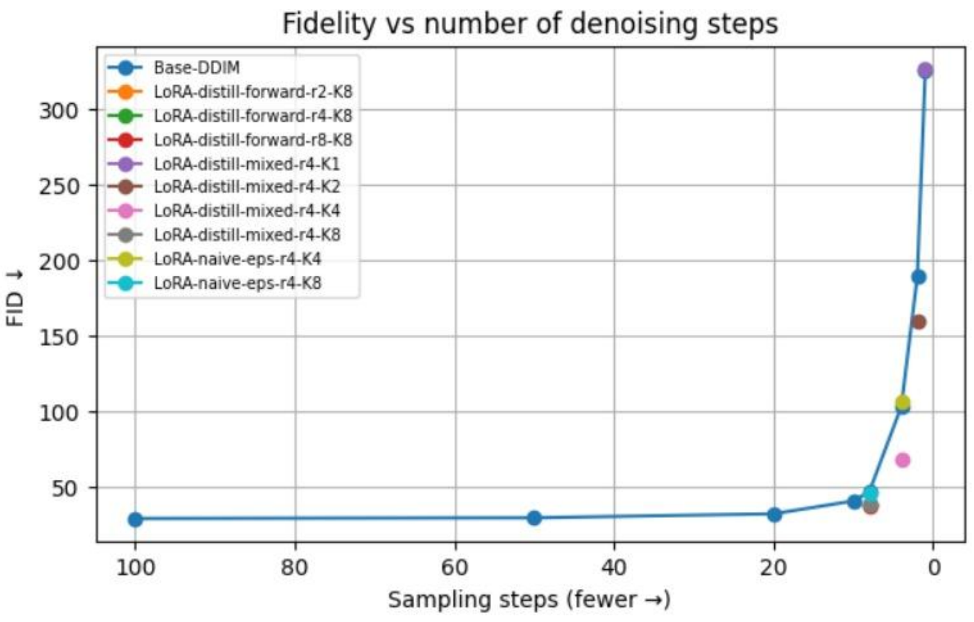
† vs. DDPM-1000. * On 1000 images (noisier).

Samples of DDIM - 100 steps.

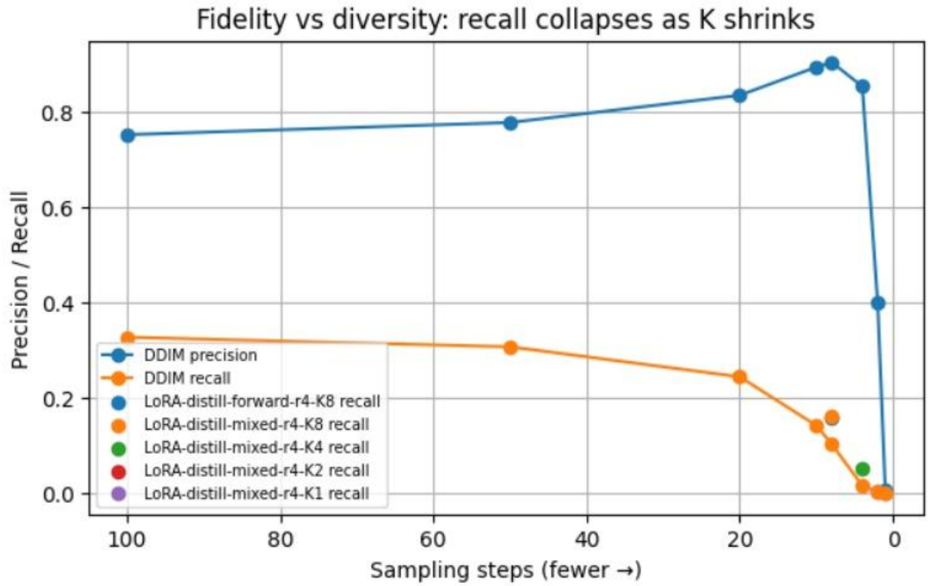
Model Evaluation – Speed vs Quality

Few steps degradation

DDIM degrades sharply below 10 steps; the distilled LoRA students sit below the DDIM curve at the same step count.



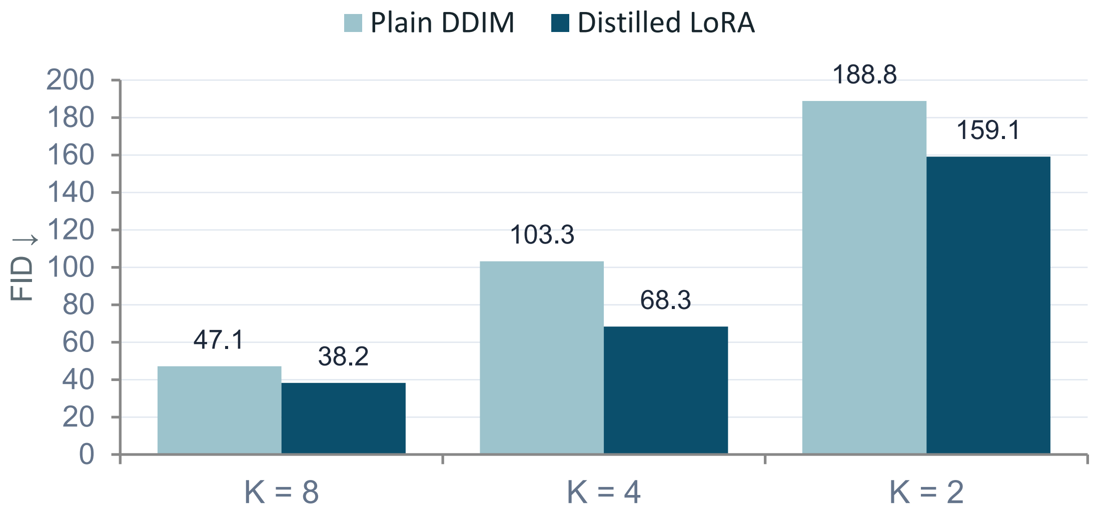
FID mixes quality and variety; Precision/Recall separate them.



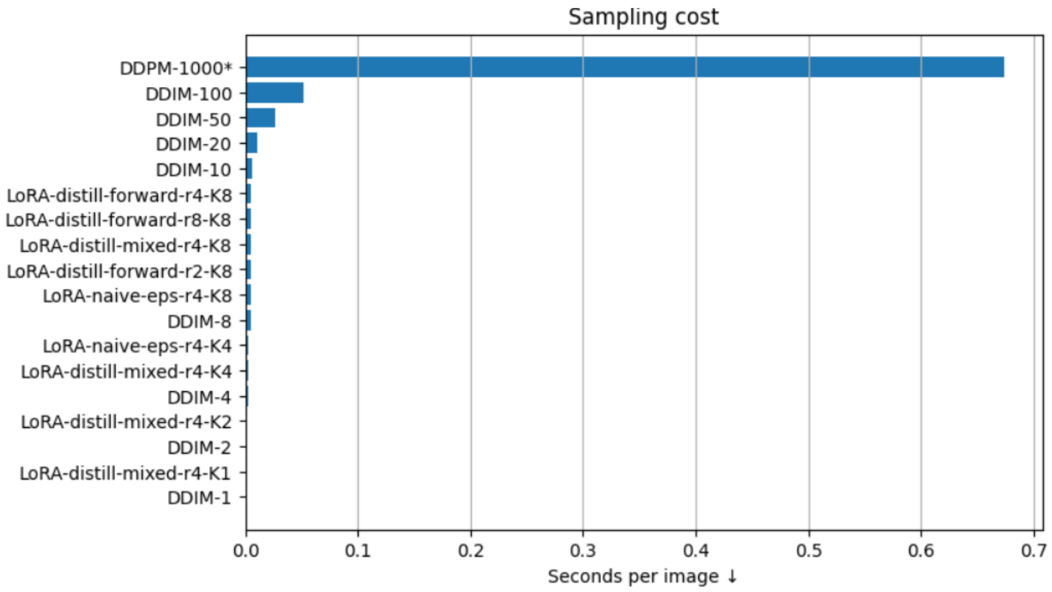
The real casualty is diversity

As steps drop 100 → 8, precision even **rises (0.75 → 0.90)** — single images stay realistic. But recall **drops (0.33 → 0.10 → 0)**: the model covers less and less of the data and produces less variety.

Model Evaluation — LoRA vs. DDIM



Same steps, same noise, same weights — only the LoRA adapter differs.



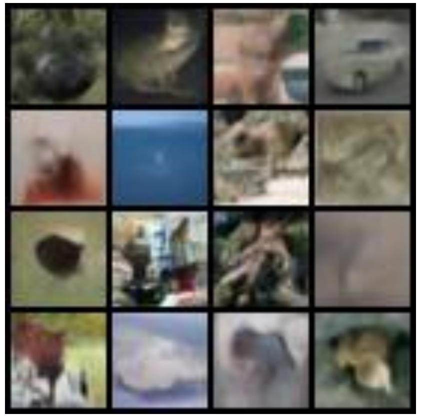
LoRA obtains at most the same execution time as DDIM at matched steps but with higher quality.

K	DDIM FID	Distilled FID	ΔFID	DDIM Recall	Distilled Recall	Time ratio
8	47.12	**38.24**	**+8.9**	0.102	0.159	1.01×
4	103.28	**68.34**	**+34.9**	0.015	0.051	1.01×
2	188.84	**159.12**	**+29.7**	0.001	0.002	1.01×
1	325.11	326.57	−1.5	0.000	0.000	1.00×

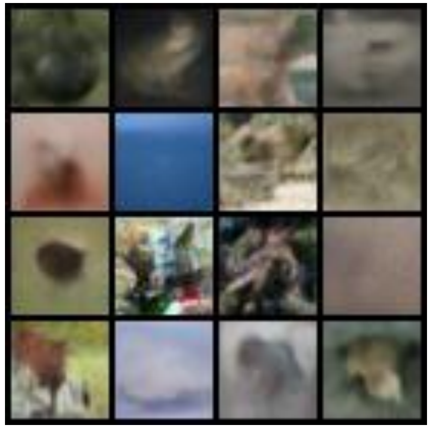
The distilled K=8 model is **10.6×** faster than DDIM-100, and K=4 is **18.9×** faster but at a fraction of DDIM's quality loss and with no inference overhead.

Model Evaluation — LoRA vs. DDIM (matched steps)

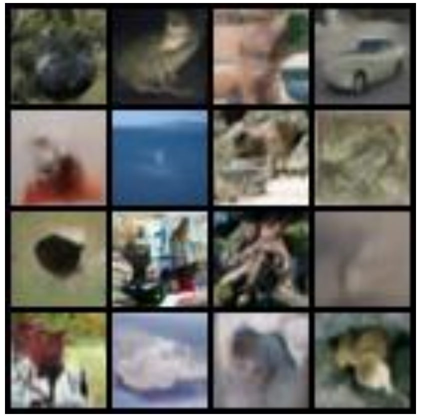
DDIM-8



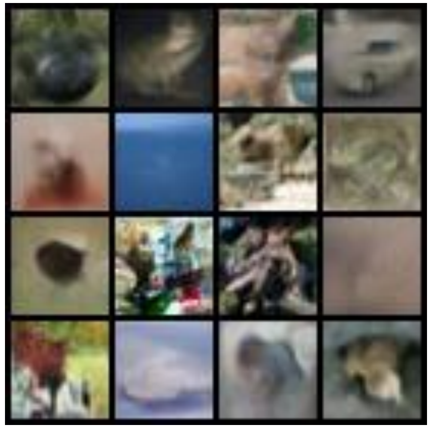
DDIM-4



DDIM-1



LoRA-k8



LoRA-k4

LoRA-k1

Model Evaluation — Ablations

Objective

Distillation vs naive ε-loss

	FID ↓
Naive-eps LoRA	46.16
Distilled LoRA	**38.24**

The gain comes from the trajectory-matching objective, not the extra parameters.

LoRA rank

r = 2 / 4 / 8

	FID ↓
rank 2	38.32
rank 4	38.48
rank 8	**37.67**

Rank barely matters: even rank 2 captures the benefit → truly parameter-efficient.

State source

forward vs mixed

	FID ↓
forward only	38.48
mixed states	**38.24**

Essentially tied: a negative result. We do not rely on trajectory-state mixing. K = 8

K = 1 fails (FID 326): a single denoising step is not enough for this adapter setup. K = 8 and K = 4 are the meaningful models.

Conclusions

We presented a parameter-efficient few-step distillation of a DDPM using LoRA adapters and a trajectory-matching objective, evaluated on CIFAR-10.

10–19× faster

than the DDIM-100 baseline, at matched quality

Better few-step FID

+8.9 for K=8 and +34.9 for K=4 over plain DDIM

Diversity restored

recall partially recovered (+0.057 for K=8)

No inference overhead

adapters merge back; only ~0.1% of weights trained

Limitations: single-step (K=1) sampling is not recovered on this baseline; absolute quality is hardware-limited (FID≈29). Confirming the smaller K=8 gain would need multiple seeds.

References

[1] J. Ho, A. Jain, P. Abbeel. **Denoising Diffusion Probabilistic Models.** *NeurIPS 2020.*

[2] J. Song, C. Meng, S. Ermon. **Denoising Diffusion Implicit Models.** *ICLR 2021.*

[3] T. Salimans, J. Ho. **Progressive Distillation for Fast Sampling of Diffusion Models.** *ICLR 2022.*

[4] E. Hu, Y. Shen, P. Wallis, et al. **LoRA: Low-Rank Adaptation of Large Language Models.** *2021.*

[5] T. Kynkäänniemi, T. Karras, S. Laine, J. Lehtinen, T. Aila. **Improved Precision and Recall Metric for Assessing Generative Models.** *NeurIPS 2019.*